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BALM: A Model-Agnostic Framework for Balanced Multimodal Learning under Imbalanced Missing Rates

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Abstract

*Learning from multiple modalities often suffers from imbalance, where information-rich modalities dominate optimization while weaker or partially missing modalities contribute less. This imbalance becomes severe in realistic settings with imbalanced missing rates (IMR), where each modality is absent with different probabilities, distorting representation learning and gradient dynamics. We revisit this issue from a training-process perspective and propose **BALM**, a model-agnostic plug-in framework to achieve balanced multimodal learning under IMR. The framework comprises two complementary modules: the Feature Calibration Module (FCM), which recalibrates unimodal features using global context to establish a shared representation basis across heterogeneous missing patterns; the Gradient Rebalancing Module (GRM), which balances learning dynamics across modalities by modulating gradient magnitudes and directions from both distributional and spatial perspectives. BALM can be seamlessly integrated into diverse backbones, including multimodal emotion recognition (MER) models, without altering their architectures. Experimental results across multiple MER benchmarks confirm that BALM consistently enhances robustness and improves performance under diverse missing and imbalance settings. Code available at: https://github.com/np4s/BALM_CVPR2026.git*

1. Introduction

Multimodal learning has achieved substantial progress by jointly exploiting complementary information from audio, visual, and textual modalities [2, 17, 26, 41]. However, real-world multimodal systems rarely operate under perfect conditions, as sensor failures, recording noise, or acquisition costs often cause partial or complete modality loss [25, 40, 47]. This phenomenon leads to *incomplete multimodal learning*, where missing data affect both representation quality and cross-modal interaction [47]. Effectively handling incomplete or missing modalities remains a central challenge for building reliable multimodal systems [18]. Beyond incompleteness, multimodal learning inherently faces *modality imbalance* [32, 39, 43], where modalities contribute unequally due to differences in signal reliability, feature granularity, and data scale. When modalities are missing, this imbalance worsens as the dominant modality overfits while the less frequent or corrupted ones receive limited supervision, leading to a biased equilibrium that weakens robustness and generalization.

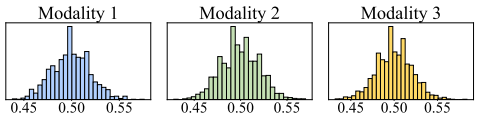


Figure 1. Modality-specific missing rates' distribution of a trimodal dataset under Share Missing Rate (SMR=0.5).

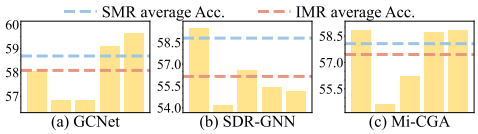


Figure 2. Model accuracy under SMR=0.5 and five equivalent IMR settings with $(r_A, r_L, r_V) \in \{0.3, 0.5, 0.7\}$ on IEMOCAP.

Recent advances in missing-modality learning have explored two main paradigms to enhance robustness under incomplete data: *alignment-based methods*, which align representations between complete and incomplete inputs using contrastive, correlation, or augmentation techniques [16, 19, 23, 25], and *generation-based methods*, which reconstruct missing modalities through autoencoders, variational inference, graph reasoning, or diffusion models [21, 34, 44, 49]. These approaches generally overlook the imbalance that arises when missingness occurs during training. Notably, some models [6, 16] compute reconstruction losses using the full data, whereas some approaches [44, 49] do not employ reconstruction but rely on pretrained encoders trained on full modalities.

Most prior studies further simplify the problem by adopting the *Shared Missing Rate (SMR)* assumption [6, 16, 49],

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where all modalities are randomly dropped with the same probability. As illustrated in Fig. 1, when $SMR{=}0.5$, each modality exhibits comparable missing proportions, resulting in a statistically balanced yet idealized setting. In contrast, realistic multimodal systems often encounter uneven data degradation across modalities due to heterogeneous acquisition or noise. We refer to this scenario as the *Imbalanced Missing Rate (IMR)* condition [27, 28, 48], which introduces persistent exposure disparity and optimization bias across modalities. As shown in Fig. 2, traditional missing-modality models exhibit unstable behavior under *IMR* settings, particularly when the missing rates across modalities differ significantly. This imbalance causes performance fluctuations, as models struggle to adapt when one modality becomes substantially less reliable than the others.

Our approach. We study multimodal learning under the *Imbalanced Missing Rate (IMR)* setting, where modality absence occurs from the beginning of training and follows distinct missing rates. IMR introduces two coupled challenges: (1) *representation imbalance*, where heterogeneous missing patterns distort unimodal feature distributions and hinder consistent cross-modal fusion [30]; and (2) *learning imbalance*, where gradients are dominated by frequently observed modalities, leading to biased convergence [7]. To address these issues, we propose **BALM** (***B**alanced **A**gnostic **L**earning under Imbalanced **M**issing Rates*), a lightweight and model-agnostic plug-in framework that mitigates both representation- and optimization-level imbalance under IMR. Our propose BALM integrates seamlessly with existing multimodal backbones, particularly those used in MER. It comprises two complementary modules: a Feature Calibration Module (`FCM`) that aligns representations across varying missing patterns, and a Gradient Rebalancing Module (`GRM`) that harmonizes optimization dynamics by adaptively modulating gradient magnitudes and directions. Our main contributions are:

- We propose **BALM**, a plug-in framework that enables robust multimodal learning under the *Imbalanced Missing Rate* condition without redesigning existing architectures.
- We design two complementary modules: a *Feature Calibration Module* (`FCM`) for representation-level alignment and a *Gradient Rebalancing Module* (`GRM`) for optimization-level balancing.
- We conduct extensive experiments on multiple MER benchmarks, showing that BALM consistently enhances robustness across standard backbones, imbalance-oriented models, and missing-modality methods.

2. Related Work

2.1. Incomplete Multimodal Learning

Recent studies on incomplete multimodal learning can be broadly categorized into two paradigms [47]. **Alignment-based methods** aim to bridge the gap between complete and incomplete inputs by aligning their latent representations. Contrastive objectives [15, 19] encourage modality-invariant embeddings, while correlation-based techniques such as canonical correlation analysis and its extensions [1, 8, 31] learn cross-modal coherence. Other efforts incorporate masking or noise-based augmentation [25, 45] to simulate missingness and improve robustness. Although effective at aligning feature spaces, these methods depend on balanced modality exposure and struggle under heterogeneous missing conditions. **Generation-based methods** explicitly reconstruct missing modalities to recover full-modality representations. Autoencoder-based frameworks [21, 29] learn deterministic mappings between observed and missing modalities, while variational inference models [33, 49] impose probabilistic constraints to improve generative consistency. Graph-based completion networks [16, 23] capture structural dependencies among modalities and instances, whereas diffusion-based approaches [4, 5, 35] leverage denoising priors to iteratively refine reconstructed signals. Despite their generative flexibility, these methods still assume identical missing probabilities across modalities and rarely consider imbalance during optimization.

2.2. Imbalanced Multimodal Learning

Imbalance in multimodal learning occurs when some modalities dominate training because they provide stronger or more reliable information [32, 39, 47]. This causes biased gradients and uneven representation learning, where dominant modalities are updated more effectively while weaker ones lag behind. Prior studies address this issue through data re-sampling [37], adaptive feature calibration [20, 47], objective regularization [20, 36, 42], and gradient modulation [12, 14, 22, 24, 38]. These approaches enhance cooperation among modalities when all are fully available. Under incomplete conditions, imbalance worsens as low-availability modalities provide fewer samples and gradients, slowing convergence and degrading representations. The interplay between modality imbalance and missing data further complicates optimization and weakens cross-modal alignment. Our work addresses these issues by jointly calibrating features and rebalancing gradients for stable learning under heterogeneous missing conditions.

3. Methodology

Given a multimodal dataset $\mathbb{D} = \{(x_i, y_i)\}_{i=1}^{N}$ with M modalities and label space $\mathcal{Y}$, the multimodal feature of sample i is denoted as $x_i = \{x_i^m\}_{m=1}^{M}$, where $x_i^m \in \mathbb{R}^{d_m}$ represents the feature of modality m. To simulate missing-modality scenarios, we introduce a stochastic masking operator $\mathcal{M}(\cdot; \mathbf{r})$ parameterized by a missing-ratio vector $\mathbf{r} = [r_1, \ldots, r_M]$, where $r_m \in [0, 1)$ denotes the probability that modality m is absent. The missing mask vector of a mul-

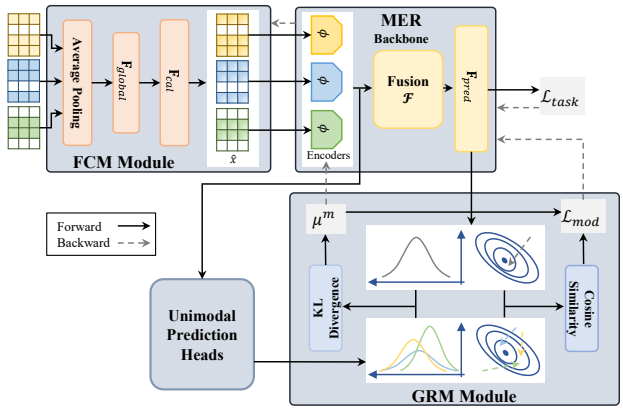


Figure 3. Overview of BALM, consisting of a Feature Calibration Module (FCM) and a Gradient Rebalancing Module (GRM) for balanced multimodal learning under imbalanced missing rates.

timodal sample is obtained via $e_i \sim \mathcal{M}(x_i; \mathbf{r})$, where each observation indicator $e_i^m \in \{0, 1\}$ indicates the availability of modality m for sample i. The indicator is generated with regards to the missing probability r_m while ensuring at least one modality is available for each sample. The incomplete dataset is defined as $\tilde{\mathbb{D}} = \{(\tilde{x}_i, y_i)\}_{i=1}^N$, where $\tilde{x}_i$ is the incomplete multimodal sample with unimodal feature $\tilde{x}_i^m = x_i^m$ if $e_i^m = 1$, and $\tilde{x}_i^m = \vec{0}$ if otherwise.

3.1. Shared- and Imbalanced Missing Rate

Shared Missing Rate (SMR). In the shared setting, all modalities follow the same Bernoulli missingness pattern, where for each sample i and modality m. The observation indicator $e_i^m \in \{0, 1\}$ is drawn via $P(e_i^m = 1) = 1 - r_{\text{shared}}$ and $P(e_i^m = 0) = r_{\text{shared}}$, with $r_{\text{shared}} \in [0, 1)$ denoting the shared missing rate. The joint masking distribution is:

$$p_{\text{SMR}}(e) = \frac{\prod_{m=1}^M (1 - r_{\text{shared}})^{e^m} r_{\text{shared}}^{1-e^m}}{1 - r_{\text{shared}}^M}, \tag{1}$$

where $e = \{e^m\}_{m=1}^M$ represents the binary masking pattern. Here, the denominator normalizes by excluding the all-missing case. For each modality m, the empirical missing ratio $\hat{r}_m = \frac{1}{N} \sum_{i=1}^N (1 - e_i^m)$ follows a Binomial distribution $B(N, r_{\text{shared}})$ with expectation and variance:

$$\mathbb{E}[\hat{r}_m] = r_{\text{shared}}, \qquad \text{Var}[\hat{r}_m] = \frac{r_{\text{shared}}(1 - r_{\text{shared}})}{N}. \tag{2}$$

As N increases, $\hat{r}_m$ converges to $\mathcal{N}\big(r_{\text{shared}}, \frac{r_{\text{shared}}(1-r_{\text{shared}})}{N}\big)$, implying all modalities are statistically equivalent (see Fig. 1), and any difference among $\hat{r}_m$ arises from stochastic fluctuation rather than systematic imbalance.

Imbalanced Missing Rate (IMR). In realistic multimodal environments, different modalities exhibit heterogeneous missing probabilities ($r_m \neq r_{m'}$) due to sensor reliability, recording noise, or acquisition cost. The corresponding masking distribution becomes modality-specific:

$$p_{\text{IMR}}(e) = \frac{\prod_{m=1}^M (1 - r_m)^{e^m} r_m^{1-e^m}}{1 - \prod_{m=1}^M r_m}, \tag{3}$$

where $p_{\text{IMR}}(e)$ and $p_{\text{SMR}}(e)$ denote the probability of observing a particular masking pattern e under IMR and SMR, respectively. Unlike the homogeneous p_{SMR}, the heterogeneous p_{IMR} assigns higher probability mass to combinations retaining low-r_m modalities. The degree of imbalance can be quantified by a divergence:

$$\Delta_{\text{IMR}} = \mathcal{D}\big(p_{\text{IMR}}(e) \,\|\, p_{\text{SMR}}(e)\big), \tag{4}$$

where $\mathcal{D}(\cdot\|\cdot)$ is a distributional divergence measure, e.g., KL or Jensen–Shannon distance. A larger Δ_{IMR} reflects stronger modality heterogeneity and sampling bias.

From Distributional to Learning Imbalance. The heterogeneous masking distribution p_{IMR} induces a persistent *distributional shift* between the training data distribution $\mathbb{P}_{\text{train}}(\tilde{x}, y)$ and the full-modality distribution $\mathbb{P}_{\text{full}}(x, y)$, whose magnitude scales with Δ_{IMR}. This imbalance further propagates to the optimization process. Given a backbone model $\mathcal{F}$ that produces prediction $\hat{y}_i = \mathcal{F}(\tilde{x}_i)$, the learning objective under IMR can be formulated as:

$$\mathcal{L}_{\text{IMR}} = \mathbb{E}_{(\tilde{X}, Y) \sim P_{\text{IMR}}} \Big[\sum_{m=1}^M (1 - r_m)\, \ell(f_m(\tilde{X}_m), Y) \Big], \tag{5}$$

where ℓ denotes the task loss for modality branch m. The gradient contribution of each modality scales with its availability ratio $(1 - r_m)$ is computed via:

$$\frac{\partial \mathcal{L}_{\text{IMR}}}{\partial \theta_m} \propto (1 - r_m)\, \mathbb{E}\left[\frac{\partial \ell(f_m(\tilde{X}_m), Y)}{\partial \theta_m} \right]. \tag{6}$$

Consequently, modalities with higher missing ratios receive weaker supervision and converge more slowly, resulting in **representation underfitting** and **optimization dominance** by low-missing-rate modalities. (See more in Suppl.B)

Motivation. The above analysis reveals two coupled challenges under IMR: (1) *uneven exposure leading to inconsistent intra-modal representations*, and (2) *gradient-level inequality during optimization*. To address these issues, we propose a model-agnostic framework that performs rebalancing at both the representation and optimization levels, as illustrated in Fig. 3.

3.2. Representation-level Rebalancing with Feature Calibration Module

IMR introduces an imbalanced exposure pattern, where some modalities are observed far more often than others.

This imbalance distorts the corresponding unimodal representations, weakening their semantic consistency and compromising cross-modal learning [28]. Here, we introduce **Feature Calibration Module (`FCM`)** to align modality-specific features that are distorted by uneven exposure under IMR. Instead of imputing missing signals, it leverages global contextual cues aggregated from available modalities to recalibrate unimodal embeddings.

Feature aggregation under IMR. Following [13], we obtain modality-wise descriptors to represent the overall statistics of available features. Given the masked features $\tilde{x}_i^m$ and binary indicators e_i^m from the masking operator $\mathcal{M}(x;\mathbf{r})$, the unimodal global descriptor is computed by averaging available features:

$$x_{glob}^m = \frac{\sum_{i=1}^N \tilde{x}_i^m}{\varepsilon + \sum_{i=1}^N e_i^m}, \tag{7}$$

where $x_{global}^m \in \mathbb{R}^{d_m}$ and ε is a smoothing hyper-parameter.

Cross-modal context extraction. The unimodal descriptors are concatenated and passed through a fully connected layer to capture shared cross-modal context:

$$x_{global} = \mathbf{ReLU}\big(\mathbf{F}_{global}([x_{glob}^{m_1}, x_{glob}^{m_2}, ..., x_{glob}^{m_M}])\big), \tag{8}$$

where $x_{global} \in \mathbb{R}^{d_{global}}$ encodes the global information jointly derived from all available modalities $\{m_j\}$. This representation is then projected to modality-specific calibration weights through independent fully connected layers:

$$w_{cal}^m = \mathbf{F}_{cal}^m(x_{global}), \tag{9}$$

where $w_{cal}^m \in \mathbb{R}^{d_m}$ represents the contextual calibration weight for modality m.

Feature calibration. Each modality feature is adaptively recalibrated through excitation-based calibration weights derived from global contextual information [10]. These weights highlight the importance of each modality and help correct representation bias under IMR. Formally, a gating operation with sigmoid activation is applied to rescale each unimodal feature:

$$\hat{x}_i^m = (1 + \sigma(w_{cal}^m)) \odot \tilde{x}_i^m, \tag{10}$$

where $\hat{x}_i^m \in \mathbb{R}^{d_m}$, $\odot$ and $\sigma(\cdot)$ denote element-wise multiplication and sigmoid function, respectively. Features of missing modalities remain as $\vec{0}$.

Although the recalibrated features $\hat{x}_i^m$ differ in composition, they preserve dimensionality and seamlessly integrated into any backbone. This shared calibration mitigates IMR-induced representation discrepancies and provides a consistent feature basis for downstream optimization.

3.3. Model-Agnostic Integration of Multimodal Emotion Recognition Backbone

For compatibility with existing Multimodal Emotion Recognition (MER) frameworks, we keep this stage unchanged. Each sample in MER corresponds to an utterance u_i with multimodal inputs $\tilde{x}_i = \{\tilde{x}_i^a, \tilde{x}_i^v, \tilde{x}_i^l\}$ and an emotion label $y_i \in \mathcal{Y}$, where a, v, l denote acoustic, visual, lexical modalities, respectively. The model predicts $\hat{y}_i$ representing the probability distribution over classes. Our plug-in modules work seamlessly with any backbone, treating missing modalities as $\vec{0}$ during training and inference.

Unimodal encoding. Each modality feature $\hat{x}_i^m$ from the Feature Calibration Module (FCM) is processed by a modality-specific encoder to obtain semantic embeddings:

$$z_i^m = \phi^m(\hat{x}_i^m), \tag{11}$$

where $z_i^m \in \mathbb{R}^{d_{emb}}$ denotes the embedding of modality m, and $\phi^m(\cdot)$ represents its encoder[1].

Multimodal fusion. The resulting unimodal embeddings are then combined through a fusion network:

$$h_i = \mathcal{F}([z_i^a, z_i^v, z_i^l]), \tag{12}$$

where $h_i \in \mathbb{R}^{d_h}$ is the fused multimodal representation, and $\mathcal{F}(\cdot)$ denotes the fusion function[2].

Prediction layer. The fused representation is mapped to the output space through a fully connected prediction head followed by a softmax layer:

$$\hat{y}_i = \delta(\mathbf{F}_{pred}(h_i)), \tag{13}$$

where $\hat{y}_i \in \mathbb{R}^{|\mathcal{Y}|}$ is the predicted class probability distribution and $\delta(\cdot)$ denotes the softmax operator.

Optimization objective. A task-specific loss $\ell(\cdot)$, typically Cross-Entropy or L1, is employed to measure the discrepancy between the predicted and ground-truth labels:

$$\mathcal{L}_{task} = \sum_{i=1}^N \ell(\hat{y}_i, y_i), \tag{14}$$

where N is the number of utterances in a mini-batch. This loss is used to optimize the parameters of the backbone via gradient-based updates:

$$\theta_{(t+1)} = \theta_{(t)} - \alpha \frac{\partial \mathcal{L}_{task}}{\partial \theta}, \tag{15}$$

where θ denotes the model parameters, t is the current training iteration, and α is the learning rate.

3.4. Optimization-level Rebalancing with Gradient Rebalancing Module

Generally, popular modalities drive most gradient updates, resulting in skewed convergence [7]. Here, we propose **Gradient Rebalancing Module (`GRM`)** to balance learning dynamics across modalities, in which each modality-specific component receives fair and consistent supervision under IMR. We use the latest unimodal embeddings as optimization targets, while their encoders act as modality-specific components.

[1] In practice, $\phi^m(\cdot)$ is typically implemented using an LSTM, GRU, or multilayer perceptron (MLP) depending on the modality type.

[2] $\mathcal{F}(\cdot)$ can take forms such as concatenation, attention, tensor fusion, or graph-based aggregation, depending on the backbone architecture.

Unimodal Prediction Heads. To quantify optimization discrepancies among modalities, we introduce lightweight unimodal prediction heads that operate independently. Each head consists of two fully connected layers applied to the embeddings from Eq. 11. The first layer projects the embedding dimension to that of the multimodal representation in Eq. 12, followed by a ReLU activation as follows:

$$h_i^m = \mathbf{ReLU}\big(\mathbf{F}_{map}^m(z_i^m)\big), \tag{16}$$

where $h_i^m \in \mathbb{R}^{d_h}$. The second layer produces modality-specific predictions using the same formulation as Eq. 13:

$$\hat{y}_i^m = \delta\big(\mathbf{F}_{pred}^m(h_i^m)\big), \tag{17}$$

where $\hat{y}_i^m \in \mathbb{R}^{|\mathcal{Y}|}$ and $\delta(\cdot)$ denotes the softmax operator. We deliberately design these two-layer classifiers to ensure that the gradient matrices of all modality-specific prediction layers $\mathbf{F}_{pred}^m$ share the same shape $(d_h \times |\mathcal{Y}|)$ as the framework's main prediction layer $\mathbf{F}_{pred}$. Each unimodal head is optimized using the same loss function $\ell(\cdot)$ as in Eq. 14, where the task-specific loss for modality m is:

$$\mathcal{L}_{task}^m = \sum_{i=1}^{N} \ell(\hat{y}_i^m, y_i), \quad \theta_{(t+1)}^m = \theta_{(t)}^m - \alpha \frac{\partial \mathcal{L}_{task}^m}{\partial \theta^m}. \tag{18}$$

where θ^m is updated via standard gradient descent.

Gradient Rebalancing Module (GRM). This module mitigates the optimization bias introduced by imbalanced missing modalities. It adjusts gradient updates of modality-specific encoders from two complementary perspectives: *distribution-driven* and *spatial-driven*.

Distribution-driven modulation. Since complete unimodal distributions are unobservable under IMR, modality contributions become skewed as dominant modalities overshadow weaker ones. We therefore approximate the multimodal prediction distribution as a stable reference capturing intra- and inter-modal information. To measure the discrepancy between unimodal predictions and the multimodal reference, we use the Kullback–Leibler (KL) divergence as:

$$\mathcal{D}_{\mathrm{KL}}^m = \sum_{i=1}^{N} \mathbf{KL}\big(\hat{y}_i^m \,\|\, \hat{y}_i\big), \tag{19}$$

where $\hat{y}_i$ and $\hat{y}_i^m$ denote the multimodal and unimodal predicted class probability distribution from Eq. 13 and Eq. 17, respectively. A larger $\mathcal{D}_{\mathrm{KL}}^m$ indicates slower convergence of modality m toward the shared multimodal distribution.

Absolute divergence values are insufficient for tracking learning dynamics, as dominant modalities stabilize quickly while others fluctuate. We thus compute the *relative learning progress* of each modality from the change in divergence between consecutive iterations:

$$\Delta_{\mathrm{KL}}^m = \begin{cases} \mathcal{D}_{\mathrm{KL}}^{m^{(t)}}, & t = 0, \\ \mathcal{D}_{\mathrm{KL}}^{m^{(t-1)}} - \mathcal{D}_{\mathrm{KL}}^{m^{(t)}}, & \text{otherwise}, \end{cases} \tag{20}$$

A larger Δ_{KL}^m indicates faster alignment with the multimodal reference. Based on these progress signals, we define a *modulation coefficient* for each modality:

$$\mu^m = \rho \frac{\sum_{m' \in \{a,v,l\}, m' \neq m} \Delta_{\mathrm{KL}}^{m'}}{\sum_{m' \in \{a,v,l\}} \Delta_{\mathrm{KL}}^{m'}}, \tag{21}$$

where ρ is a saturation hyperparameter controlling the modulation intensity. A smaller μ^m indicates that modality m is learning faster (larger Δ_{KL}^m), and thus its gradient update should be attenuated; conversely, slower modalities receive stronger updates. Accordingly, the parameters of unimodal encoders are updated as:

$$\theta_{(t+1)}^{\phi^m} = \theta_{(t)}^{\phi^m} - \alpha\, \mu^m \frac{\partial \mathcal{L}_{\mathrm{task}}}{\partial \theta_{(t)}^{\phi^m}}, \tag{22}$$

where θ^{ϕ^m} denotes the parameters of encoder ϕ^m.

Spatial-driven modulation. Previous work [32, 39, 47] shows that multimodal optimization follows the dominant modality. This imbalance worsens under missing modalities, so we reorient gradients toward a balanced multimodal direction to stabilize training. Directly computing full unimodal gradients is computationally expensive, and comparing them against the entire model's gradients is often infeasible. Following [7], we approximate the overall model and modality-specific gradients using those of the multimodal and unimodal prediction heads, respectively:

$$\nabla_{pred} = \nabla_{\theta^{\mathbf{F}_{pred}}} \mathcal{L}_{task} = \frac{\partial \mathcal{L}_{task}}{\partial \theta^{\mathbf{F}_{pred}}}, \tag{23}$$

$$\nabla_{pred}^m = \nabla_{\theta^{\mathbf{F}_{pred}^m}} \mathcal{L}_{task}^m = \frac{\partial \mathcal{L}_{task}^m}{\partial \theta^{\mathbf{F}_{pred}^m}}, \tag{24}$$

where ∇_{pred} represents the gradient of the main multimodal classifier and ∇_{pred}^m corresponds to the gradient of modality m's prediction head. We measure the spatial discrepancy between these gradients via cosine similarity:

$$\mathcal{D}_{\cos}^m = \cos_{sim}\big(\nabla_{pred}^m, \nabla_{pred}\big), \tag{25}$$

A lower cosine similarity indicates that modality m is optimized in a direction less aligned with the multimodal objective. To avoid over-constraint, we adopt a soft spatial alignment strategy, which minimizes the weighted absolute deviation of cosine similarities:

$$\mathcal{L}_{mod} = \sum_{m \in \{a,v,l\}} \big| \mathcal{D}_{\cos}^m \times \mu^m \big|, \tag{26}$$

where μ^m are dynamic modulation coefficients from Eq. 21, adaptively weighting each modality's contribution during spatial alignment under IMR.

Table 1. Performance on IEMOCAP under different IMR configurations.

Model	(0.3,0.5,0.7)		(0.3,0.7,0.5)		(0.5,0.3,0.7)		(0.5,0.7,0.3)		(0.7,0.3,0.5)		(0.7,0.5,0.3)	
	Acc.	w-F1	Acc.	w-F1	Acc.	w-F1	Acc.	w-F1	Acc.	w-F1	Acc.	w-F1
Group 1 (Models addressing missing or imbalanced modalities)												
MMIN [49]	55.97	49.92	55.87	49.83	56.40	50.28	55.87	49.75	55.87	49.85	55.83	49.73
SDR-GNN [6]	59.46	59.34	54.16	53.59	58.53	58.42	56.56	56.77	55.14	55.22	55.45	55.17
Mi-CGA [23]	58.84	58.80	54.65	54.61	60.50	60.25	56.19	56.11	58.84	58.63	58.72	58.22
MoMKE [44]	55.39	54.19	50.83	49.70	60.18	59.94	50.77	50.15	58.26	58.45	54.78	54.92
GCNet [16]	58.04	58.36	56.81	56.83	61.61	62.06	56.81	56.99	59.64	59.78	59.09	59.19
Ada2I [22]	61.98	60.68	61.98	61.46	61.37	60.45	58.10	58.22	61.74	60.90	57.73	54.76
RedCore [28]	46.27	46.32	43.38	43.40	50.15	47.92	43.69	43.60	50.59	50.53	46.09	46.21
MCE [48]	46.52	45.54	44.36	43.48	50.65	49.22	42.64	41.70	50.34	49.45	44.73	44.40
GCNet$_{+\text{BALM}}$	61.31$_{\pm 0.75}$	61.33$_{\pm 0.70}$	57.36$_{\pm 0.76}$	57.00$_{\pm 0.83}$	61.31$_{\pm 0.63}$	61.62$_{\pm 0.59}$	57.64$_{\pm 1.10}$	57.64$_{\pm 1.22}$	59.33$_{\pm 0.46}$	59.39$_{\pm 0.50}$	60.26$_{\pm 0.83}$	60.58$_{\pm 0.96}$
Δ (vs origin GCNet)	*+3.27*	*+2.97*	*+0.55*	*+0.17*	*-0.30*	*-0.44*	*+0.83*	*+0.65*	*-0.31*	*-0.39*	*+1.17*	*+1.39*
Δ_1 (vs Grp1 mean)	*+6.00*	*+7.19*	*+4.61*	*+5.39*	*+4.21*	*+5.55*	*+5.06*	*+5.98*	*+3.03*	*+4.04*	*+6.21*	*+7.76*
Group 2 (Typical MER Models)												
MMGCN [11]	63.59	63.62	59.09	58.99	64.63	64.95	57.79	57.99	64.14	64.27	60.63	60.85
MMGCN$_{+\text{BALM}}$	64.39$_{\pm 0.88}$	63.82$_{\pm 0.89}$	60.07$_{\pm 0.70}$	60.04$_{\pm 0.65}$	65.37$_{\pm 0.61}$	65.03$_{\pm 0.63}$	62.29$_{\pm 0.75}$	62.35$_{\pm 0.70}$	65.00$_{\pm 1.54}$	64.98$_{\pm 1.77}$	61.74$_{\pm 0.96}$	61.20$_{\pm 0.82}$
Δ (vs origin MMGCN)	*+0.80*	*+0.20*	*+0.98*	*+1.05*	*+0.74*	*+0.08*	*+4.50*	*+4.36*	*+0.86*	*+0.71*	*+1.11*	*+0.35*
Δ_1 (vs Group1 mean)	*+9.08*	*+9.68*	*+7.32*	*+8.43*	*+7.95*	*+8.96*	*+9.71*	*+10.69*	*+8.70*	*+9.63*	*+7.69*	*+8.38*
MMDFN [9]	63.34	63.00	60.38	60.01	67.34	67.40	57.92	56.25	66.48	66.67	63.65	64.00
MMDFN$_{+\text{BALM}}$	67.34$_{\pm 0.84}$	66.64$_{\pm 0.94}$	63.40$_{\pm 0.45}$	63.11$_{\pm 0.47}$	68.76$_{\pm 0.22}$	68.25$_{\pm 0.18}$	64.02$_{\pm 1.11}$	63.76$_{\pm 1.05}$	67.22$_{\pm 1.03}$	67.02$_{\pm 1.06}$	65.50$_{\pm 0.47}$	64.98$_{\pm 0.64}$
Δ (vs origin MMDFN)	*+4.00*	*+3.64*	*+3.02*	*+3.10*	*+1.42*	*+0.85*	*+6.10*	*+7.51*	*+0.74*	*+0.35*	*+1.85*	*+0.98*
Δ_1 (vs Group1 mean)	*+12.03*	*+12.50*	*+10.65*	*+11.50*	*+11.34*	*+12.18*	*+11.44*	*+12.10*	*+10.92*	*+11.67*	*+11.45*	*+12.16*

Note. (r_A, r_L, r_V) denotes the missing rates for Audio, Language, and Visual modalities.
Δ and Δ_1 denote performance differences relative to the original backbone and mean of Group 1 models, respectively.

Overall Objective. The overall training objective integrates the task loss and the modulation loss as:

$$\mathcal{L} = \mathcal{L}_{task} + \tau\, \mathcal{L}_{mod}, \tag{27}$$

where τ is a trade-off hyperparameter.

The modality-specific classifiers introduced in Sec. 3.4 are optimized independently using $\mathcal{L}^m_{task}$ as auxiliary objectives and are therefore excluded from Eq. 27. The complete training procedure is summarized in Suppl.A.

4. Experiments

4.1. Experimental Settings

Datasets and Evaluation Metrics. We conduct experiments on two multimodal emotion recognition datasets: **IEMOCAP** [3] (*6-way*) and **CMU-MOSEI** [46], each containing audio (A), visual (V), and lexical (L) modalities. We adopt Accuracy (*Acc.*) and Weighted F1-score (*w-F1*) as evaluation metrics to assess model performance.

Baselines. We evaluate BALM with two main groups of baselines: (1) **Models addressing missing or imbalanced modalities**, including *MMIN* [49], *SDR-GNN* [6], *Mi-CGA* [23], *MoMKE* [44], and *GCNet* [16] for missing-modality learning; *Ada2I* [22] for modality imbalance; *RedCore* [28] and *MCE* [48] which jointly handles both aspects; and (2) **Typical MER backbones**, *MMGCN* [11] and *MMDFN* [9], representing standard multimodal architectures. All models are trained from scratch, with missing modalities simulated under shared (*SMR*) or imbalanced (*IMR*) rates defined in Sec. 3.1.

Our plug-in (**+BALM**) is integrated without altering backbone architectures and supports both early- and late-fusion setups. This setup enables evaluating BALM across incomplete, imbalanced, and standard MER backbones. Further dataset statistics, baseline descriptions, and implementation details are provided in the Suppl.E.

4.2. Main Results

Performance Comparison. Tables 1 and 2 present results under different IMR settings with $(r_A, r_L, r_V) \in \{0.3, 0.5, 0.7\}$, where integrating **BALM** consistently improves all metrics across all backbones.

As shown in Table 1, integrating BALM consistently boosts both missing-modality and emotion-recognition baselines. For example, GCNet$_{+\text{BALM}}$ yields a gain of up to $+3.27\%$ Acc over the original GCNet on the $(0.3, 0.5, 0.7)$ setting. It consistently outperforms other missing-modality baselines. Under the $(0.7, 0.5, 0.3)$ configuration, it improves w-F1 by roughly $+7.0\%$ compared with the *Group 1* mean. These results confirm its effectiveness in reducing bias caused by uneven modality availability. Compared with recent IMR-oriented works of *Group 1*, which explicitly model missing patterns, stronger fusion backbones integrated with BALM, notably MMDFN$_{+\text{BALM}}$ and MMGCN$_{+\text{BALM}}$, achieve even higher performance across different missing scenarios. In addition, BALM elevates both MER models in all missing configurations (up to 7.51% w-F1 gap over the original of MMDFN$_{+\text{BALM}}$), and raises their performance lower-bound to above 60% for both Acc and w-F1. These results suggest that rebalancing, representation- and optimization-wise, provides complementary benefits for high-capacity fusion architectures. Ada2I, which addresses imbalance learning for MER, performs better than conventional missing-modality models but still degrades under IMR, showing that gradient balancing alone is insufficient when missingness occurs.

Likewise, on CMU-MOSEI (Table 2), BALM consistently improves performance across IMR settings with smaller but steady gains, reflecting the dataset's lower modality noise and weaker sensitivity to missing-rate variations. Persistent with the observation in the previous dataset, high language-missing rates in CMU-MOSEI lead

Table 2. Performance on CMU-MOSEI under different IMR configurations.

Model	(0.3, 0.5, 0.7)		(0.3, 0.7, 0.5)		(0.5, 0.3, 0.7)		(0.5, 0.7, 0.3)		(0.7, 0.3, 0.5)		(0.7,0.5,0.3)	
	Acc.	w-F1	Acc.	w-F1	Acc.	w-F1	Acc.	w-F1	Acc.	w-F1	Acc.	w-F1
Group 1 (Models addressing missing or imbalanced modalities)												
MMIN [49]	75.87	75.52	71.93	71.60	79.72	79.69	71.46	71.30	78.92	78.45	75.65	75.81
SDR-GNN [6]	80.13	80.14	77.38	77.38	83.35	83.03	76.69	76.75	83.54	83.44	81.67	81.21
Mi-CGA [23]	81.04	80.97	77.38	77.11	82.55	82.37	76.42	76.45	82.86	82.64	80.65	80.33
MoMKE [44]	78.43	78.39	74.16	74.19	82.17	81.86	74.38	73.93	82.11	81.79	77.27	76.77
GCNet [16]	81.43	81.07	78.23	78.14	82.64	82.78	78.34	78.51	83.76	83.57	81.45	81.24
Ada2I [22]	82.42	82.28	78.07	77.53	83.82	83.57	77.68	76.97	83.38	83.49	81.87	81.61
RedCore [28]	74.88	72.89	70.51	64.25	75.05	72.82	75.01	72.79	74.37	72.71	74.94	73.59
GCNet$_{+\text{BALM}}$	81.54$_{\pm 0.58}$	81.40$_{\pm 0.59}$	78.51$_{\pm 0.61}$	78.33$_{\pm 0.49}$	83.27$_{\pm 0.26}$	83.24$_{\pm 0.33}$	79.50$_{\pm 0.99}$	79.22$_{\pm 0.84}$	83.63$_{\pm 0.35}$	83.57$_{\pm 0.48}$	82.09$_{\pm 0.75}$	81.84$_{\pm 0.72}$
Δ *(vs origin GCNet)*	*+0.11*	*+0.33*	*+0.28*	*+0.19*	*+0.63*	*+0.46*	*+1.16*	*+0.71*	*-0.13*	*+0.00*	*+0.64*	*+0.60*
Δ_1 *(vs Group1 mean)*	*+2.37*	*+2.65*	*+3.13*	*+4.02*	*+1.94*	*+2.37*	*+3.79*	*+3.98*	*+2.35*	*+2.70*	*+3.02*	*+3.19*
Group 2 (Typical MER Models)												
MMGCN [11]	80.82	80.20	78.32	77.35	84.76	84.60	78.12	78.17	84.04	84.04	81.54	81.62
MMGCN$_{+\text{BALM}}$	82.03$_{\pm 0.31}$	81.74$_{\pm 0.28}$	78.73$_{\pm 0.27}$	78.33$_{\pm 0.31}$	84.54$_{\pm 0.32}$	84.34$_{\pm 0.29}$	79.83$_{\pm 0.49}$	79.44$_{\pm 0.37}$	84.84$_{\pm 0.29}$	84.70$_{\pm 0.32}$	82.50$_{\pm 0.49}$	82.36$_{\pm 0.54}$
Δ *(vs origin MMGCN)*	*+1.21*	*+1.54*	*+0.41*	*+0.98*	*-0.22*	*-0.26*	*+1.71*	*+1.27*	*+0.80*	*+0.66*	*+0.96*	*+0.74*
Δ_1 *(vs Group1 mean)*	*+2.86*	*+2.99*	*+3.35*	*+4.02*	*+3.21*	*+3.47*	*+4.12*	*+4.20*	*+3.56*	*+3.83*	*+3.43*	*+3.71*
MMDFN [9]	82.17	81.76	78.37	78.14	84.18	83.92	78.87	79.03	84.37	84.30	81.89	81.69
MMDFN$_{+\text{BALM}}$	82.06$_{\pm 0.43}$	81.73$_{\pm 0.45}$	78.81$_{\pm 0.58}$	78.50$_{\pm 0.63}$	84.34$_{\pm 0.44}$	84.14$_{\pm 0.39}$	79.36$_{\pm 0.39}$	79.19$_{\pm 0.39}$	84.78$_{\pm 0.21}$	84.68$_{\pm 0.26}$	82.53$_{\pm 0.35}$	82.29$_{\pm 0.28}$
Δ *(vs origin MMDFN)*	*-0.11*	*-0.03*	*+0.44*	*+0.36*	*+0.16*	*+0.22*	*+0.49*	*+0.16*	*+0.41*	*+0.38*	*+0.64*	*+0.60*
Δ_1 *(vs Group1 mean)*	*+2.89*	*+2.98*	*+3.43*	*+4.19*	*+2.63*	*+3.03*	*+3.65*	*+3.95*	*+3.50*	*+3.81*	*+3.46*	*+3.64*

Note. (r_A, r_L, r_V) denotes the missing rates for Audio, Language, and Visual modalities.
Δ and Δ_1 denote performance differences relative to the original model and the mean of Group 1 models, respectively.

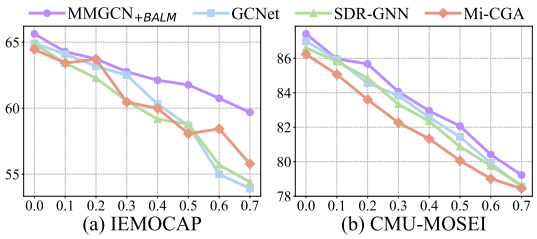


Figure 4. Average accuracy under different SMR settings

to higher performance degradation, and in return, such configurations highlight the effectiveness of BALM in elevating backbones' performance. Specifically, MMGCN$_{+\text{BALM}}$ achieves a $+1.71\%$ Acc gain over the original MMGCN, widen the gap with *Group 1* mean to $+4.12\%$ Acc under $(0.5, 0.7, 0.3)$ setting, a similar boost can also be found on GCNet$_{+\text{BALM}}$ under the same setting.

For both dataset, moderate to high loss of visual (e.g., $(0.7, 0.3, 0.5)$, $(0.5, 0.3, 0.7)$) usually leads to lower degradation comparing to other modalities. Whereas, high missing rates in language (e.g., $(0.3, 0.7, 0.5)$) lead to profound performance drops, confirming that lexical information is the most discriminative yet vulnerable modality. When audio missingness is also high alongside language (e.g., $(0.5, 0.7, 0.3)$), the degradation becomes more pronounced, underscoring the critical role of robust auditory cues. These divergent behaviors when modalities take turn suffering low/high missing rate, along with the consistent gains of the backbones after integrating with BALM, further validate the need to address modalities' discrepancy stem from IMR.

Evaluating Robustness under SMR Settings. To ensure a fair cross-evaluation, we adopt the same settings and model configurations as in the IMR experiments, while uniformly applying identical missing rates to all modalities (SMR). As shown in Fig. 4, integrating **BALM** consistently outperforms all baselines under SMR settings,[3] indicating that our approach not only handles the challenging IMR scenario but also maintains superior robustness and generalization under balanced missing conditions.

Table 3. Average performance on different modality combinations.

Setting	MMGCN		+FCM	
	Acc	w-F1	Acc	w-F1
IEMOCAP	48.09	44.81	**50.99**$_{(\uparrow 2.90)}$	**47.62**$_{(\uparrow 2.81)}$
CMU-MOSEI	75.23	69.17	**75.97**$_{(\uparrow 0.74)}$	**70.09**$_{(\uparrow 0.92)}$

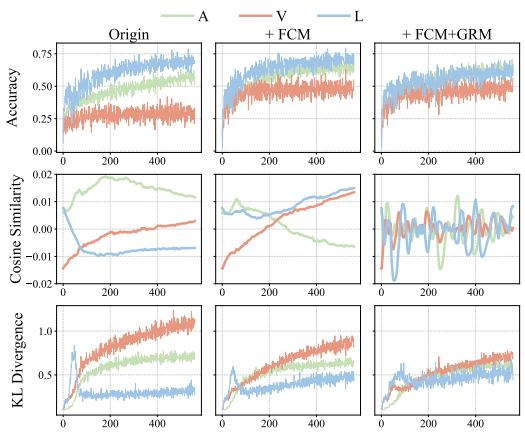


Figure 5. Modality discrepancies on IEMOCAP with MMGCN.

4.3. Ablation Study

Effect of Rebalancing Modules. Table 3 summarizes the average performance over modality combinations under the IMR setting $(r_A, r_L, r_V) = (0.5, 0.7, 0.3)$. `FCM` consistently improves results on both datasets, boosting w-F1 by $+2.81\%$ on IEMOCAP and $+0.92\%$ on MOSEI.

As illustrated in Fig. 5, `FCM` effectively enhances feature-level consistency by reducing representation dis-

[3]Each result is averaged over five independent runs with different random seeds for each shared missing rate.

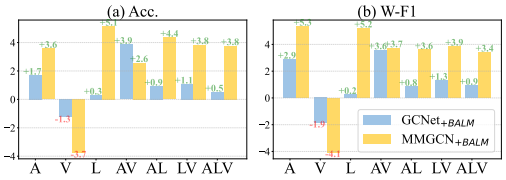


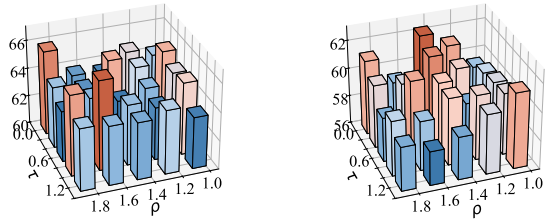
Figure 6. Improvement of GCNet$_{+BALM}$ and MMGCN$_{+BALM}$ comparing to GCNet on IEMOCAP

crepancies among modalities; however, it does not fully resolve the imbalance in gradient magnitudes across branches. Building on this, GRM further rebalances the gradient flow. In particular, the lexical branch tends to dominate optimization, resulting in large cross-modal divergence. With GRM, the inter-modal cosine similarity increases and the KL divergence stabilizes, indicating that gradient rebalancing harmonizes multimodal optimization and alleviates dominance from high-availability modalities.

Modality Combination. Fig. 6 shows that BALM consistently improves both GCNet and MMGCN across most modality combinations on IEMOCAP, with the largest gains on audio and lexical branches (up to $+5.3\%$ w-F1). Slight drops in settings where only the visual modality is available stem partly from the low quality of visual cues in IEMOCAP and partly from the fusion architecture of each baseline. Fusion-based methods like MMGCN were originally developed for MER and use modality-specific encoders and fusion mechanisms, which makes them more sensitive to variations in modality quality. As a result, they experience larger performance drops. In contrast, GCNet incorporates mechanisms to handle missing modalities and is more alignment-aware, enabling it to capture modality-specific cues and suffer less degradation under the same conditions.

Hyperparameter Sensitivity under Contrasting Missing Rates. Fig. 7 compares w-F1 of MMGCN$_{+\text{BALM}}$ under two contrasting missing-rate settings on IEMOCAP: *Config–A:* $(0.5, 0.3, 0.7)$ and *Config–B:* $(0.5, 0.7, 0.3)$. In *Config–A*, where the least missing-sensitive modality, i.e. visual, is highly missing, w-F1 is stable around $63 - 64\%$ despite the change in τ, ρ. Conversely, in *Config–B*, where the more vulnerable modality, i.e. lexical, suffers highest missing rate, the imbalance-aware regularization yields more distinct effectiveness, as higher performance can be achieved with proper selection of ρ or τ. These results underscore the importance of **adaptive regulation** for imbalanced missing-modality scenarios.

Severe Missing Rate Evaluation. Table 4 presents the robustness evaluation under extreme missing-modality conditions. As audio and language are informative modalities in MER, their absence typically leads to notable degradation. Nevertheless, BALM consistently stabilizes performance across settings. For instance, when 90% of the language modality is missing, MMGCN$_{+\text{BALM}}$ achieves a $+4.78\%$ accuracy gain on IEMOCAP compared to the mean of baselines (eg. GCNet, MMGCN). Under heavy audio missing $(0.9, 0.6, 0.6)$, it still improves by $+2.72\%$ on accuracy. Although a few settings exhibit minor performance drops, the overall trend indicates BALM effectively mitigates imbalance effects and maintains robustness even when informative modalities are severely incomplete.



(a) *Config-A*: $(0.5, 0.3, 0.7)$ (b) *Config-B*: $(0.5, 0.7, 0.3)$

Figure 7. Weighted-F1 sensitivity to hyperparameters under two missing-rate configurations of audio, language and visual.

Table 4. Performance of MMGCN with BALM under severe missing-modality settings, where the arrows indicate the relative change compared to the mean performance of the two baselines.

(r_A, r_L, r_V)	Model	IEMOCAP		CMU-MOSEI	
		Acc	w-F1	Acc	w-F1
$(0.1, 0.1, 0.9)$	GCNet	63.71	63.43	84.62	84.64
	MMGCN	68.02	67.69	85.91	85.52
	MMGCN$_{+\text{BALM}}$	**65.87**$_{\uparrow 0.01}$	**65.57**$_{\uparrow 0.01}$	**85.91**$_{\uparrow 0.64}$	**85.70**$_{\uparrow 0.62}$
$(0.1, 0.9, 0.1)$	GCNet	55.14	55.09	72.15	71.69
	MMGCN	52.74	52.69	71.55	71.70
	MMGCN$_{+\text{BALM}}$	**58.72**$_{\uparrow 4.78}$	**58.11**$_{\uparrow 4.22}$	**72.62**$_{\uparrow 0.77}$	**72.15**$_{\uparrow 0.46}$
$(0.9, 0.1, 0.1)$	GCNet	59.40	58.83	86.24	86.10
	MMGCN	64.82	64.93	86.43	86.42
	MMGCN$_{+\text{BALM}}$	**66.24**$_{\uparrow 4.13}$	**66.27**$_{\uparrow 4.39}$	**86.35**$_{\uparrow 0.01}$	**86.38**$_{\uparrow 0.12}$
$(0.6, 0.6, 0.9)$	GCNet	56.93	56.54	81.81	81.86
	MMGCN	60.63	60.75	81.32	81.16
	MMGCN$_{+\text{BALM}}$	**61.74**$_{\uparrow 2.96}$	**61.92**$_{\uparrow 3.28}$	**81.65**$_{\uparrow 0.09}$	**81.62**$_{\uparrow 0.11}$
$(0.6, 0.9, 0.6)$	GCNet	54.84	54.85	76.00	76.02
	MMGCN	55.64	55.61	76.36	75.97
	MMGCN$_{+\text{BALM}}$	**57.30**$_{\uparrow 2.06}$	**57.48**$_{\uparrow 2.25}$	**76.55**$_{\uparrow 0.37}$	75.30$_{\downarrow 0.70}$
$(0.9, 0.6, 0.6)$	GCNet	57.98	58.16	81.45	81.06
	MMGCN	59.95	59.89	81.59	81.50
	MMGCN$_{+\text{BALM}}$	**61.68**$_{\uparrow 2.72}$	**61.63**$_{\uparrow 2.61}$	**82.00**$_{\uparrow 0.48}$	**81.55**$_{\uparrow 0.27}$

5. Conclusion

This paper presents BALM, a lightweight and model-agnostic framework for multimodal learning under imbalanced missing conditions. By combining the Feature Calibration Module (FCM) and Gradient Rebalancing Module (GRM), BALM effectively mitigates both representation and optimization imbalance. Extensive experiments on multiple MER backbones demonstrate that BALM consistently improves performance across standard, incomplete, and imbalanced-modality settings. The results highlight its robustness to heterogeneous missing patterns and its ability to maintain stable optimization even when modality availability varies significantly. Future work will explore broader multimodal tasks and large-scale pretrained backbones.

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